

Deep Learning Models for Lumbar Spinal Stenosis on MRI: Model Comparison and Clinical Benchmarking

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Inventory Category:	DL stenosis / multi-pathology
Comparability / Priority:	Very High Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Architecture comparison / clinician benchmark Lumbar MRI including axial T2
Dataset & Cohort:	nan (Sample: nan)
Disease / Target:	Lumbar spinal stenosis
Model / AI Method:	CNN and transformer model comparison
Evaluation Metrics:	Accuracy/recall and clinician comparison
Key Finding / Relevance:	nan
Thesis Context (Ch 2):	Particularly relevant because the MSc thesis frames its contribution as CNN-versus-transformer comparison.

Abstract / Summary

PurposeTo compare deep learning models of different architecture for automated lumbar spinal stenosis classification on MRI and benchmark their performance against radiologists and orthopedists.MethodsLumbar spine MRI studies from Sep-2015 to Sep-2019 were retrospectively obtained. Exclusion criteria included previous spinal instrumentation, suboptimal image quality, post-gadolinium studies, and severe scoliosis. Axial T2-weighted and sagittal T1-weighted images were used. Studies were split into training/validation and test sets. An external test set of 100 studies was used. Training data were labelled by 4 radiologists using predefined gradings. Two models, CNN-based and transformer-based, were developed. Consensus labelling by two expert spine radiologists served as the reference standard. Test sets were labelled by 8 participants (2 general radiologists, 2 radiologists-in-training, 2 orthopedists, 2 orthopedists-in-training). Detection recall (%), interrater agreement (Gwet κ), sensitivity, and specificity were evaluated.Results564 MRI lumbar spines were included (mean age = 52 $\pm$ 19[SD]; 302 women), with 464(82%) and 100(18%) for training/validation and internal testing, respectively. Both models showed high recall for all regions of interest (> 94%), similar to participants. Dichotomous classification (normal/mild vs. moderate/severe) by the CNN model, transformer model, and participants showed respective kappas for central canal 0.99/0.99/0.97-0.98, lateral recesses 0.98/0.94/0.81-0.94, and neural foramina 0.98/0.95/0.91-0.95 on internal testing (p ConclusionThe CNN model showed superior performance, and the transformer model showed similar to superior performance compared to clinicians for classifying lumbar spinal stenosis. These models could assist clinicians in report generation, surgical planning and education.